SIGN-LANGUAGE

Spring 2026

Meeting No.5

System Development

& Datasets

SYSTEM UPDATES

Model num_frames: 32
Model image_size: 224
10 labels for testing were selected

Time Window: 3 seconds (previously it was 4.5 seconds)
So, the sampling interval is: 3/32 = 0.094 sec
more validation videos for training (15 to be exact)

```
Model num_frames: 32
Model image_size: 224
Labels: ['a', 'a lot', 'abdomen', 'able', 'about', 'above', 'accent', 'accept', 'accident', 'accomplish']
```

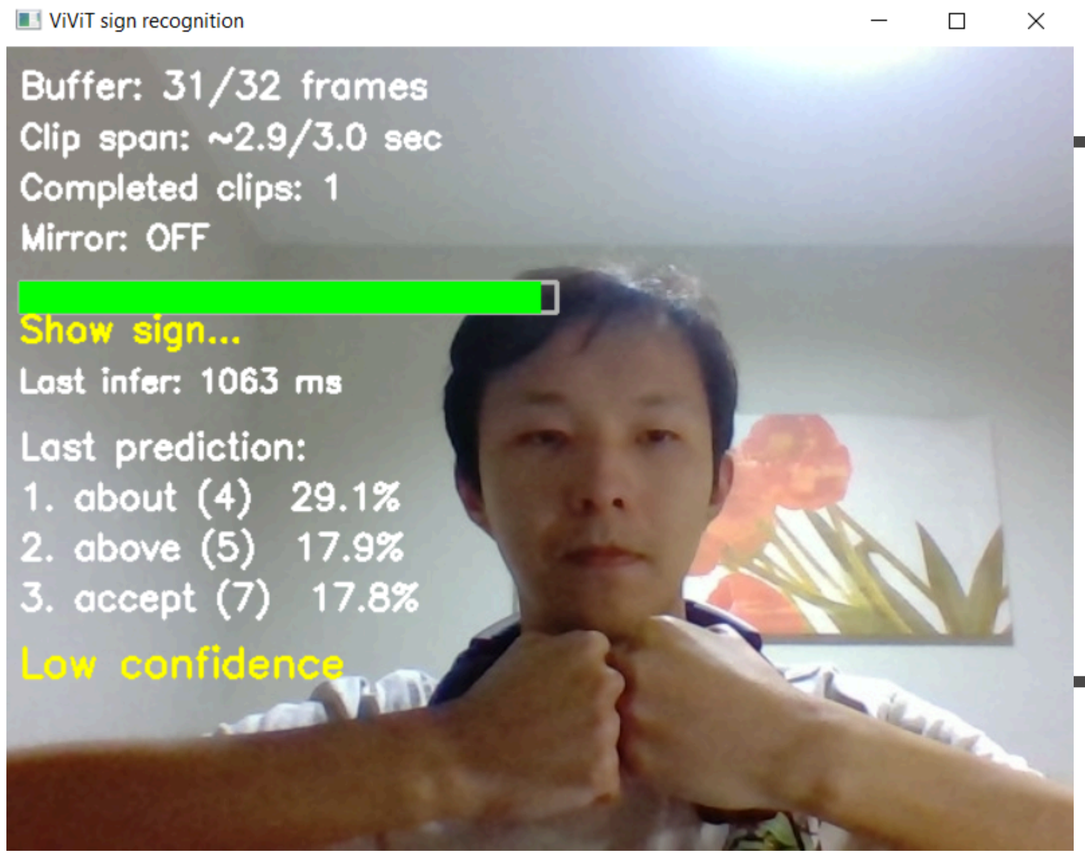
SYSTEM UPDATES (CONT)

Different videos in ONE class (showed the different sign) were deleted

Epoch 01 | train_loss=1.8239 train_acc=47.7% | val_loss=2.4146 val_acc=6.7%
Epoch 02 | train_loss=1.2392 train_acc=77.3% | val_loss=2.4639 val_acc=6.7%
Epoch 03 | train_loss=0.9511 train_acc=93.2% | val_loss=2.4432 val_acc=20.0%
Epoch 04 | train_loss=0.6693 train_acc=95.5% | val_loss=2.4017 val_acc=20.0%
Epoch 05 | train_loss=0.5209 train_acc=97.7% | val_loss=2.3976 val_acc=26.7%
Epoch 06 | train_loss=0.3956 train_acc=97.7% | val_loss=2.3995 val_acc=26.7%

However, label mapping is fixed now
Also, confidence filtering was added

reset buffer, mirroring, pause functions were added



SYSTEM UPDATES (CONT)

Case 1:
accident: 85%
others: very low
model is confused between multiple classes

Case 2:
accident: 22%
able: 19%
about: 17%
...

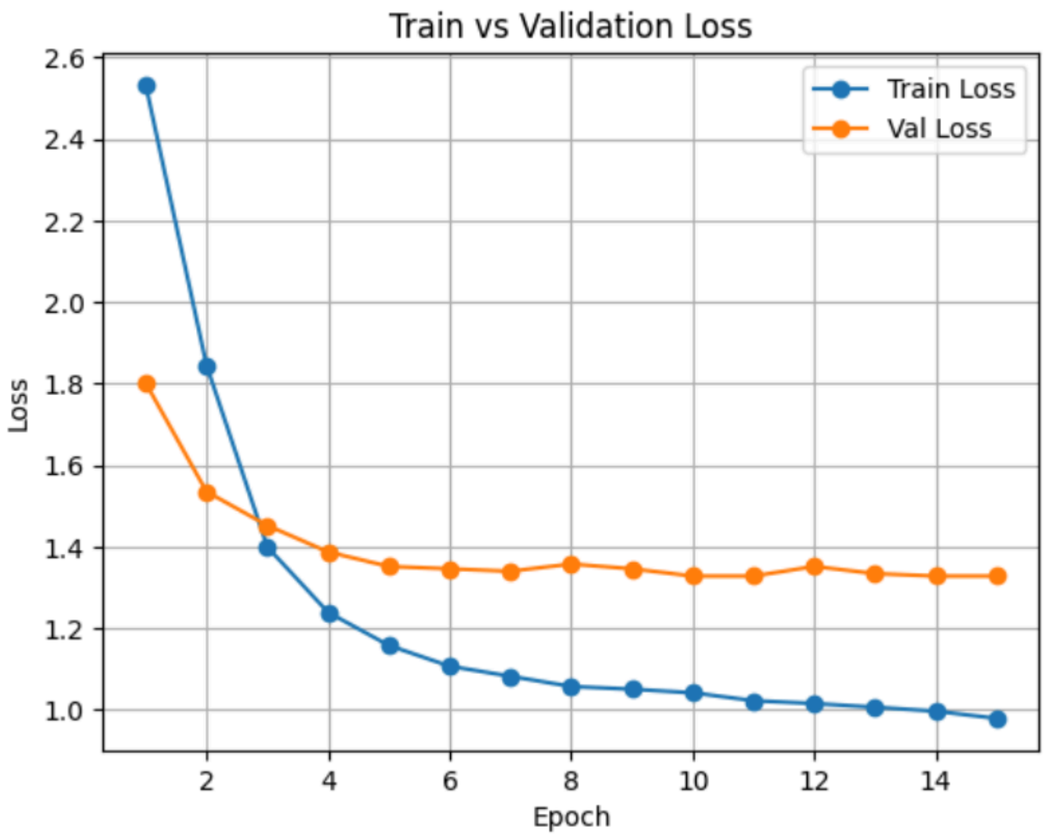
model is confused between multiple classes

CONF_THRESHOLD = 0.35

if it's below, then it shows label "Low confidence"

UPDATED RESULTS
ViViT with LoRA fine-tuning for WLASL-100

Number of epochs: 15
Total training time: 10714.3s
Per-epoch training time: 714.3s
Test accuracy: 50.0%



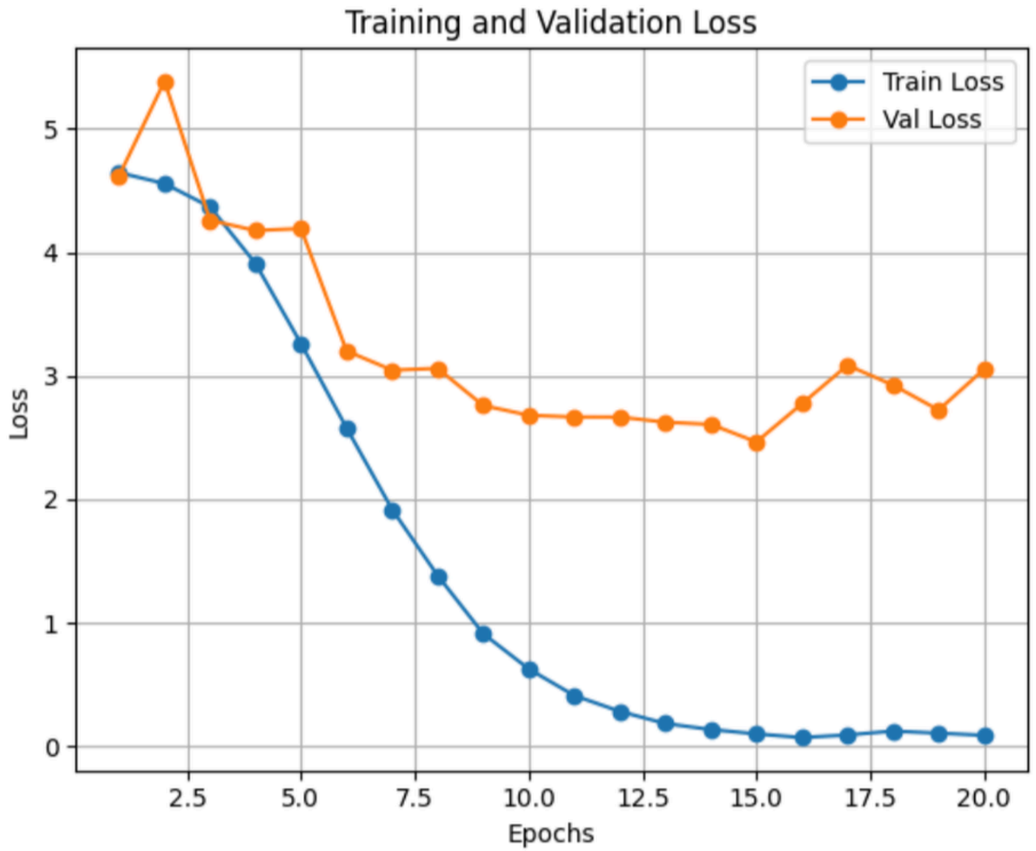
UPDATED RESULTS
ViViT with LoRA fine-tuning for WLASL-300

Number of epochs: 70
Total training time: 125902.3s
Per-epoch training time: 1798.6s
Test accuracy: 6.3%

SlowFast for WLASL_100 with uniform sampling

Total training time: 5912.32 s
Training time per epoch: 295.62 s
Final test accuracy: 49.59 %

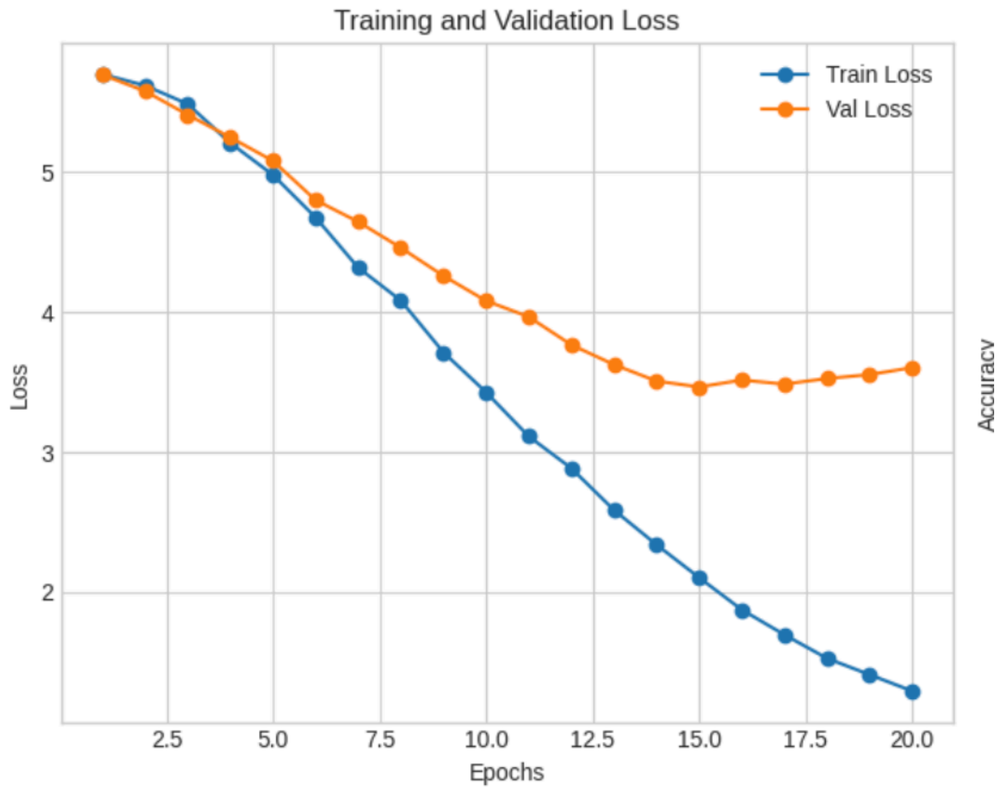
Test Accuracy for SlowFast with random sampling: 54.26%



SlowFast for WLASL_300 with uniform sampling

Total training time: 15053.45 s
Training time per epoch: 751.45 s
Final test accuracy: 35,69 %

Test Accuracy for SlowFast with random sampling: 38.27%



GitHub Link:

AbdollaZ/seniorproject_signlanguage: This is a repository for hosting CSCI 409 Senior Project…

This is a repository for hosting CSCI 409 Senior Project documentation and files. Project topic is Sign Language…

GitHub

THANK YOU